12 Patrick has to work out the exact value of $64^{\frac{1}{4}}$

Patrick says,

"$\frac{1}{4}$ of 64 is 16 so $64^{\frac{1}{4}} = 16$"

Explain what is wrong with what Patrick says.

64^(1/4) means the fourth root of 64.

64^(1/4) = 2√2 ≈ 2.828, not 16.

1 mark

Exponent 1/4 means fourth root.

(Total for Question 12 is 1 mark)